What gets interpreted where? Scope, binding, Condition C in Japanese scrambling chains

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1 INTRO

A moved DP can be interpreted below its surface position (*reconstruction*) for both scope (1) and variable binding (2):

(1) a. [$_{DP}$ How many books] did Beccy decide to read __? *decide* > *many*
b. [$_{DP}$ A student from Storrs] is likely __ to study Japanese syntax. *likely* > ∃

(2) a. [$_{DP}$ Which of his$_1$ students] did every professor$_1$ talk to __? (Fox 1999: 172)
b. [$_{DP}$ His$_1$ father] seems to every boy$_1$ __ to be a genius. (Fox 1999: 161)

Given the copy theory of movement, such reconstruction effects are often thought to be due to interpreting a lower DP-copy in the chain (Chomsky 1993; Romero 1998; Fox 1999, 2000, a.o.):

(3) a. SS: [$_{XP}$ DP$_k$ … ZP … [$_{YP}$ DP$_k$ …]]
b. LF: [$_{XP}$ DP$_k$ … ZP … [$_{YP}$ DP$_k$ …]] ZP > DP

Importantly, on these approaches, where a lower DP-copy is interpreted in LF, reconstruction for scope/binding is predicted to systematically interact with Condition C.

Scope trapping. Condition C can force a wide-scope interpretation of the DP, when a narrow-scope interpretation would cause a Condition C violation. In (4a) coreference forces wide scope of the DP; with name and pronoun switched, both readings return (4b). Similarly, in (5a) the predicate *invent* forces the *wh*-item to scope low. But this triggers Condition C, and thus, (5a) is bad. In (5b), the *wh*-item can scope high, and thus, no Condition C effect. Hence, scope reconstruction and Condition C correlate (Heycock 1995; Romero 1998; Fox 1999).

(4) a. [$_{DP}$ A student of Beccy$_i$'s] seems to her$_i$ __ to be highly intelligent. ✓ ∃ > *seem*; ✗ *seem* > ∃
b. [$_{DP}$ A student of hers$_i$] seems to Beccy$_i$ __ to be highly intelligent. ✓ ∃ > *seem*; ✓ *seem* > ∃

(5) a. *[$_{DP}$ How many stories about Beccy$_i$] is she$_i$ likely to invent __? low scope forced
b. [$_{DP}$ How many stories about Beccy$_i$] is she$_i$ really upset by __? wide scope possible

Binding. Binding into the moved DP from below the coreferring pronoun triggers a Condition C effect (6a). Binding from above doesn't (6b).

(6) a. *[$_{DP}$ Which (of the) paper(s) [$_{RC}$ that he$_1$ gave to Ms. Brown$_j$]] did she$_j$ hope __ that every student$_1$ will revise __?
b. [$_{DP}$ Which (of the) paper(s) [$_{RC}$ that he$_1$ gave to Ms. Brown$_j$]] did every student$_1$ hope __ that she$_j$ will read __? (Fox 1999: 174)

The above facts naturally fall out from an account where reconstruction is interpreting a lower DP-copy in LF. In the above cases, reconstruction brings the R-expression with it and triggers Condition C:

(7) a. [$_{DP}$ … R-exp$_i$ …] … pron$_i$ … [$_{DP}$ … R-exp$_i$ …] no reconstruction; Condition C ✓
b. *[$_{DP}$ … R-exp$_i$ …] … pron$_i$ … [$_{DP}$ … R-exp$_i$ …] scope/binding reconstruction; Condition C ✗

Today: Look at some Japanese scrambling data that complicates this tidy picture. Specifically, look at some cases where scope, binding, and Condition C effects come apart:

(8) a. [$_{DP}$ D R] … Y … [$_{DP}$ D R]

b. $[_{DP}$ D R] … Y … $[_{DP}$ D R]

While this talk is mostly exploring the data, I will also outline a tentative analysis. Building on previous approaches (Takahashi and Hulsey 2009; Gong 2022), I will propose that these effects can be accounted for with a version of *wholesale late merger*. On this analysis, the full DP is not present in every copy position, but rather is built up throughout the derivation.

Warning: I want to emphasize that this analysis is *tentative*. While I think something of this shape is on the right track, this analysis needs serious refinements. Minimally though, I believe that it achieves empirical adequacy.

2 BACKGROUND: JAPANESE SCRAMBLING

In Japanese, the DO can scramble across the IO (9a), it can scramble over the subject (9b), and it can scramble out of an embedded clause (9c) (Kuno 1973; Hoji 1985; Saito 1985, 1992, 2003; Tada 1993; Miyagawa 1997, a.m.o.).[1]

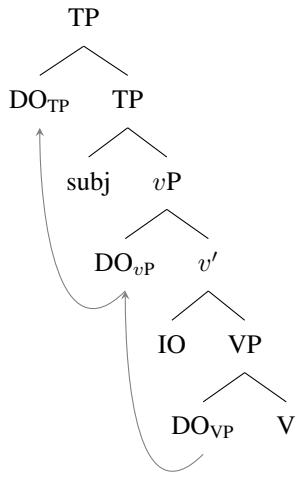
(9) a. Taroo-wa Hanako-o dareka-ni __ syookaisita.
Taro-TOP Hanako-ACC someone-DAT introduced
'Taro introduced Hanako to someone.' VPS

b. Hanako-o Taroo-wa __ suisensita.
Hanako-ACC Taro-TOP recommended
'Taro recommended Hanako.' SS

c. Hanako-o Taroo-wa [$_{CP}$ Jiroo-ga __ suisensita to] omotteiru.
Hanako-ACC Taro-TOP Jiro-NOM recommended C think
'Taro thinks Jiro recommended Hanako.' LS

Following Saito (2003, 2015), I'm going to assume that VPS involves movement to vP and SS involves movement from vP to TP. For today, we're going to ignore long scrambling and just look at clause-internal scrambling. So a scrambled DO can have up to three copies: in VP (DO_{VP}), at the vP edge (DO_{vP}), and in TP (DO_{TP}).

(10)



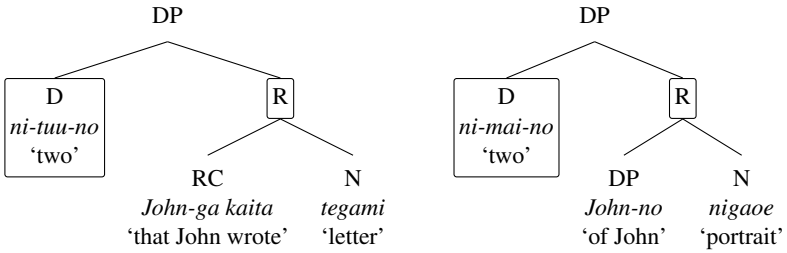
3 THREE GENERALIZATIONS

Before going over the data, a quick diagnostic point. In much of the data below, we probe scope, binding, and Condition C independently of each other. So we need a way to talk about the parts of a DP **separately**. I assume a DP has two parts:

[1] Unless cited, all data come from M. Maeda, G. Ciferri Muramatsu, and Y. Yagi.

- a **D**(eterminer): the source of the DP's quantificational force, including numerals and quantifiers;
- a **R**(estrictor): everything in D's complement, i.e. the NP and its modifiers and complements (RCs, genitive arguments).

(11)



Scope tells us where the D is interpreted relative to other scopal elements, i.e. wide/narrow; *binding* into the DP and *Condition C*, which are checked on the LF structure, tell us where R sits in that structure. Specifically, is the R c-commanded by the binder or co-refering element in LF.

3.1 DO SCRAMBLED TO vP CAN RECONSTRUCT ONLY FOR SCOPE

When the DO scrambles above the IO to vP, it can reconstruct for scope (12a), (12b).

(12) a. John-wa [$_{DO}$ dareka-o] dono sensee-ni-mo __ syookaisita.
John-TOP someone-ACC which teacher-DAT-MO introduced
'John introduced someone to every teacher.' ✓ $\forall > \exists$ (Yagi 2026)

b. John-wa [$_{DO}$[$_{RC}$ Tookyoo-ni funinsuru] futa-ri-no sensee]-o dono gakkoo-ni-mo __
John-TOP Tokyo-DAT be.assigned two-CL-GEN teacher-ACC which school-DAT-MO
syookaisita.
introduced
'John introduced two teachers assigned to Tokyo to every school.' ✓ $\forall > 2$

While a scrambled DO in vP can reconstruct below the IO for scope, it can't for binding (13a) vs. (13b) (Yagi 2026).

(13) a. John-wa [$_{IO}$ dono gakkoo-ni-mo$_1$] [$_{DO}$[$_{RC}$ soko$_1$-ni funinsuru] sensee]-o syookaisita.
John-TOP which school-DAT-MO there-DAT be.assigned teacher-ACC introduced
'John introduced to every school$_1$ a teacher assigned there$_1$.' ✓ bound

b.??John-wa [$_{DO}$[$_{RC}$ soko$_1$-ni funinsuru] sensee]-o [$_{IO}$ dono gakkoo-ni-mo$_1$] __ syookaisita.
John-TOP there-DAT be.assigned teacher-ACC which school-DAT-MO introduced
Int. 'John introduced to every school$_1$ a teacher assigned there$_1$.' ✗ bound (Yagi 2026)

Further, scrambling over the IO neutralizes Condition C effects (14b) vs. (14a):

(14) a. *Mary-ga kare$_i$-ni [$_{DO}$ John$_i$-no nigaoe]-o miseta (koto)
Mary-NOM he-DAT John-GEN portrait-ACC showed fact
'Mary showed him a portrait of John.'

b. Mary-ga [$_{DO}$ John$_i$-no nigaoe]-o kare$_i$-ni __ miseta (koto)
Mary-NOM John-GEN portrait-ACC he-DAT showed fact
'Mary showed him a portrait of John.'

Interestingly, scope trapping effects are not observed when a DO with an R-expression scrambles over a pronoun IO. As shown in (15b), the DO can scope below the IO despite containing an R-expression coreferential with the pronoun IO.

(15) **Context:** $\forall > \exists$: John introduced Ken to his friend Amy, Lee to his friend Ben, and Mia to her friend Cat (each a friend of the group Ken, Lee and Mia).

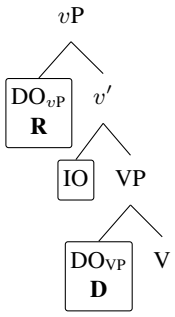
a. *John-wa karera$_i$ minna-ni [$_{DO}$ [Ken-to Lee-to Mia]$_i$-no tomodati]-o syookaisita.
John-TOP they all-DAT Ken-and Lee-and Mia-GEN friend-ACC introduced
'John introduced them$_i$ all to a friend of Ken, Lee and Mia$_i$.' ✗ Condition C

b. John-wa [$_{DO}$ [Ken-to Lee-to Mia]$_i$-no tomodati]-o karera$_i$ minna-ni __ syookaisita.
John-TOP Ken-and Lee-and Mia-GEN friend-ACC they all-DAT introduced
'John introduced a friend of Ken, Lee and Mia$_i$ to them$_i$ all.' ✓ Condition C + $\forall > \exists$

Given the above facts, a scrambled DO in vP can reconstruct below the IO for scope but not for binding. Further, Condition C effects are generally neutralized and there's no scope trapping (cf. English (5), (4)). Thus, we have the generalization in (16):

(16) **G1:** A DO scrambled to vP can take scope below the IO, but its R is never c-commanded by the IO at LF.

(17)



3.2 SCOPE TRAPPING WHEN SCRAMBLING ABOVE THE SUBJECT

When a DO scrambles over the subject to TP, it can reconstruct below the subject for scope (18) (Hoji 1985; Tada 1993):

(18) Daremo-o dareka-ga __ suisensita.
everyone-ACC someone-NOM recommended
'Someone recommended everyone.' ✓ $\forall > \exists$; ✓ $\exists > \forall$

Further, scrambling the DO over the subject can neutralize Condition C effects between the subject and the DO, as in (19a) vs. (19b), (19c) (Saito 1985; Tada 1993).

(19) a. *Kare$_i$-ga [$_{DO}$ John$_i$-no nigaoe]-o mita (koto)
he-NOM John-GEN portrait-ACC saw fact
'He saw a portrait of John.'

b. [$_{DO}$ John$_i$-no nigaoe]-o kare-zisin$_i$-ga __ mita (koto)
John-GEN portrait-ACC he-self-NOM saw fact
'A portrait of John, he himself saw.'

c. [$_{DO}$ ni-mai-no John$_i$-no nigaoe]-o kare-zisin$_i$-ga __ mita (koto)
two-CL-GEN John-GEN portrait-ACC he-self-NOM saw fact
'Two portraits of John, he himself saw.'

Interestingly, unlike our cases of scrambling to vP, over the IO, when the DO has scrambled to TP, we can observe scope trapping effects.

As shown in (20), the scrambled DOs can reconstruct for scope below the IO here:

(20) a. [$_{DO}$[$_{RC}$ John-ga kaita] ni-tuu-no tegami]-o Mary-ga dono gakusei-ni-mo __ miseta (koto)
John-NOM wrote two-CL-GEN letter-ACC Mary-NOM which student-DAT-MO showed fact
'Mary showed every student two letters that John wrote.' ✓ $2 > \forall$; ✓ $\forall > 2$

b. [$_{DO}$ ni-mai-no John-no nigaoe]-o Mary-ga dono gakusei-ni-mo __ miseta (koto)
two-CL-GEN John-GEN portrait-ACC Mary-NOM which student-DAT-MO showed fact
'Mary showed every student two portraits of John.' ✓ $2 > \forall$; ✓ $\forall > 2$

It can also reconstruct below the subject but above the IO, as in (21):

(21) **Context:** $\forall > 3 > \exists$: teacher A introduced students 1, 2 and 3 to Ken, Lee and Mia, respectively; teacher B introduced students 4, 5 and 6 to Ned, Oli and Pat, respectively; and so on for every teacher.

a. [$_{DO}$ San-nin-no gakusei]-o dono sensee-mo dareka-ni __ syookaisita.
three-CL-GEN student-ACC which teacher-MO someone-DAT introduced
'Every teacher introduced three students to someone.' ✓ $\forall > 3 > \exists$

b. [$_{DO}$ San-nin-no gakusei]-o dono sensee-mo dareka hitori-ni __ syookaisita.
three-CL-GEN student-ACC which teacher-MO someone one.CL-DAT introduced
'Every teacher introduced three students to some one person.' ✓ $\forall > 3 > \exists$

But when the subject is a pronoun coreferential with an R-expression in the scrambled DO, the DO must scope high:

(22) **Scenarios:** $2 > \forall$: the same two letters/portraits for every student; $\forall > 2$: a different pair for each.

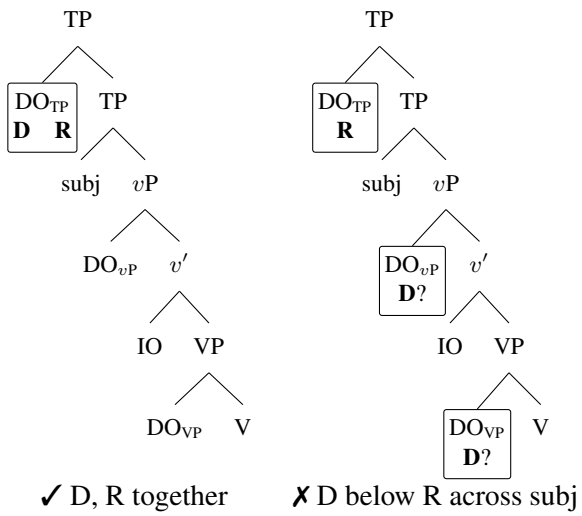
a. [$_{DO}$[$_{RC}$ John$_i$-ga kaita] ni-tuu-no tegami]-o kare$_i$-ga dono gakusei-ni-mo __ miseta (koto)
John-NOM wrote two-CL-GEN letter-ACC he-NOM which student-DAT-MO showed fact
'He showed every student two letters that John wrote.' ✓ $2 > \forall$; ✗ $\forall > 2$

b. [$_{DO}$ ni-mai-no John$_i$-no nigaoe]-o kare$_i$-ga dono gakusei-ni-mo __ miseta (koto)
two-CL-GEN John-GEN portrait-ACC he-NOM which student-DAT-MO showed fact
'He showed every student two portraits of John.' ✓ $2 > \forall$; ✗ $\forall > 2$

In this way, scrambling over the subject can create scope trapping configurations (like English). In (22), if the DO's R is interpreted below the subject, a Condition C violation is triggered. So the DO has to be interpreted high. This is different from our cases of scrambling to vP, over the IO (15b), in those cases there was no scope trapping but here there is.

(23) **G2:** When a DO is scrambled to TP, if its R must be above the subject at LF, the DO can't take scope below the subject.

(24)



✓ D, R together ✗ D below R across subj

Interestingly, the relation between D and R isn't symmetric. When the DO is scrambled to TP, D can never take scope below the position of R (G2). But R can be below D: there are cases where the DO scopes above the subject but is still bound by it.

As shown in (25), (26), the DO is scrambled to TP and can have a wide-scope interpretation relative to the subject. Further, the DO is bound by the subject.

(25) [$_{DO}$[$_{RC}$ Soitu$_1$-ga kai-ta] aru ronbun]-o [$_S$ san-nin-ika-no kooho-sya-ga]$_1$
that.guy-NOM write-PST certain paper-ACC three-CL-or.fewer-GEN candidate-person-NOM
__ teisyutu-si-ta.
submission-do-PST
'A certain paper he$_1$ wrote, at most three candidates$_1$ submitted.' ✓ bound; *a paper > at most three*

(26) [$_{DO}$[$_{RC}$ Soitu$_1$-ga kai-ta] ni-hon-no ronbun]-o [$_S$ san-nin-ika-no
that.guy-NOM write-PST two-CL-GEN paper-ACC three-CL-or.fewer-GEN
kooho-sya-ga]$_1$ __ teisyutu-si-ta.
candidate-person-NOM submission-do-PST
'Two papers he$_1$ wrote, at most three candidates$_1$ submitted.' ✓ bound; 2 > *at most three*

Caveat: binding **alone** doesn't decisively show that the Rs in (25), (26) are interpreted below the subject at LF. It has been argued that in Japanese, variable binding, like anaphor binding, can be satisfied at any point in the derivation (Saito 2003). So the DO could be bound low and then move to TP, where it is interpreted in LF.

However, there's a way to check whether binding really puts R low: combine binding with a Condition C configuration. If binding requires R to be interpreted below the binder, then any pronoun *above* the binder should trigger Condition C with an R-expression inside R.

Yagi (2026) runs exactly this test with long scrambling. In (27), the DO contains both a bound pronoun, *soko*, bound by the embedded subject *dono gakkoo-mo*, and an R-expression, *Hanako*. The matrix subject *kanojo* is above the binder.

- In base order (27a), *kanojo* c-commands *Hanako*, so coreference is out, as expected.
- After scrambling (27b), the DO is above *kanojo* on the surface. But the bound reading of *soko* still blocks coreference: binding puts R below *dono gakkoo-mo*, and therefore below *kanojo*, so Condition C is triggered.
- With name and pronoun switched (27c), binding and coreference are fine together: R is still low, but now nothing c-commands the name.

(27) a. *Kanojo$_i$-wa [$_{CP}$ dono gakkoo-mo$_1$ [$_{DO}$[$_{RC}$ Hanako$_i$-ga soko$_1$-ni suisen-sita] sensee]-o
she-TOP which school-MO Hanako-NOM there-DAT recommend-PST teacher-ACC
saiyoo-suru to] omotteiru.
hire C think
'She$_i$ thinks every school$_1$ hires a teacher Hanako$_i$ recommended there$_1$.' coreference

b. *[$_{DO}$[$_{RC}$ Hanako$_i$-ga soko$_1$-ni suisen-sita] sensee]-o kanojo$_i$-wa [$_{CP}$ dono gakkoo-mo$_1$
Hanako-NOM there-DAT recommend-PST teacher-ACC she-TOP which school-MO
__ saiyoo-suru to] omotteiru.
hire C think
'She$_i$ thinks every school$_1$ hires a teacher Hanako$_i$ recommended there$_1$.' ✗ bound + coreference

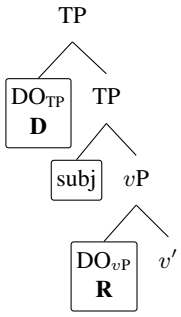
c. [$_{DO}$[$_{RC}$ kanojo$_i$-ga soko$_1$-ni suisen-sita] sensee]-o Hanako$_i$-wa [$_{CP}$ dono gakkoo-mo$_1$
she-NOM there-DAT recommend-PST teacher-ACC Hanako-TOP which school-MO
__ saiyoo-suru to] omotteiru.
hire C think
'Hanako$_i$ thinks every school$_1$ hires a teacher she$_i$ recommended there$_1$.' ✓ bound + coreference
(Yagi 2026: (15b), (16))

So binding into a scrambled DO tracks where R is interpreted: if it were satisfied before movement, (27b) should allow both readings at once. If the same holds for clause-internal scrambling, (25) and (26) show R below the subject and D above it.

Returning to (25), (26), where the scrambled DO is bound by the subject but scopes over it, we can make the following generalization:

(28) **G3:** When a DO scrambled to TP, it can take scope above the subject while its R is c-commanded by the subject at LF.

(29)



3.3 PUTTING IT ALL TOGETHER

Unlike English, it looks like scope, binding, and Condition C come apart in Japanese scrambling. However, not in a completely unrestricted way.

We saw that a scrambled DO's R is never interpreted below the vP edge (G1). The DO's D can be interpreted below R only when the DO undergoes scrambling to vP. If the DO scrambles to TP, above the subject, D is trapped as high as R (G2), although R can still be interpreted below D (G3).

The question: What fixes the position of R, and why can D be interpreted below R when the DO is in vP but not when it is in TP?

4 ANALYSIS

Recall the three generalizations:

- **G1:** A DO scrambled to vP: D can be interpreted below the IO; R never is.
- **G2:** A DO scrambled to TP: if R is interpreted above the subject, D is too.
- **G3:** A DO scrambled to TP: R can be interpreted below the subject while D is above it.

What we need is a way to interpret movement chains that can account for G1–G3.

I'm going to argue that standard approaches to reconstruction, i.e. lower XP-copy interpretation or semantic reconstruction, can't do the job.

Instead, I'm going to suggest that G1–G3 can be captured on an approach where the scrambled DP is built in stages across the movement chain. In particular, I'm going to develop an account of Japanese scrambling in terms of *wholesale late merger* (Takahashi and Hulsey 2009). The reason that the D and R of the scrambled DP can be interpreted in different positions is that D and R are present in different positions in the structure.

4.1 ALTERNATIVES

Syntactic reconstruction. In LF, a mechanism selects one copy of the moved XP for interpretation; selecting a lower copy gives reconstruction (Chomsky 1993; Romero 1998; Fox 1999, 2000):

(30) $[_{TP}\ DP_k \ldots\ YP \ldots\ [_{VP}\ DP_k \ldots]]$ YP > D, R

Scope, binding, and Condition C are all evaluated on the same copy, so D and R always go together.

Problem: this gets G2, but not G1 or G3. A DO in vP can scope below the IO (12a) without being bound by it (13b); and a DO in TP can scope above the subject while being bound by it (25).

Semantic reconstruction. A lower copy is converted into a higher-type trace (Cresti 1995; Rullmann 1995; Yagi 2026). The DP stays in its surface position in the LF structure, so R is never c-commanded by anything below it (no Condition C, no binding from below). But the DP's meaning is fed into the lower position, so its quantificational force, i.e. its scope, is low:

(31) a. LF: $[_{TP}\ DP_k\ [\ \lambda\mathcal{G}_k \ldots\ YP \ldots\ [_{VP}\ \mathcal{G}_k \ldots]]]$ $\mathcal{G}_k$: $\langle\langle e,t\rangle,t\rangle$

b. $[\![\lambda\mathcal{G}_k\,[\ldots YP \ldots \mathcal{G}_k \ldots]]\!]([\![DP]\!]) = [\ldots [\![YP]\!] \ldots [\![DP]\!] \ldots]$ YP > D; R > YP

Problem: this gets G1, but not G2 or G3. First, if semantic reconstruction is always available, D could always scope low, so there'd be no scope trapping (22b). Second, on this approach R always stays high, so it can never be interpreted lower than D (25).

What we need is a mechanism that (i) lets R be interpreted in some lower copies, but never inside VP (G1, G3); and (ii) lets D be interpreted below R only inside vP (G1, G2).

I'm going to argue that a version of *wholesale late merger* (Takahashi and Hulsey 2009) with some plausible independent assumptions can do the job.

4.2 WHOLESALE LATE MERGER OF RESTRICTORS

There's pretty compelling evidence that certain elements can be merged counter-cyclically (Lebeaux 1988, 1991; Nissenbaum 1998; Ochi 1999; Fox 1999, 2002; Stepanov 2001a,b; Bošković 2004; Takahashi and Hulsey 2009; Gong 2022).

Important for us: restrictors of a DP, i.e. RCs and complements of N, can be merged counter-cyclically.

RCs (Lebeaux 1988, 1991). Under *wh*-movement, an R-expression in an RC escapes Condition C (32a); when it's in the complement of N, it can't (32b). In the base, VP-internal position, both RCs and complements of N always trigger a Condition C effect (33).

(32) a. $[_{DP}$ What claim $[_{RC}$ that Beccy$_i$ made]] did she$_i$ believe __? RC (adjunct)

b. *$[_{DP}$ Which inspection of Beccy$_i$'s office] did she$_i$ resent __? complement of N

(33) a. *She$_i$ believed $[_{DP}$ the claim $[_{RC}$ that Beccy$_i$ made]].

b. *She$_i$ resented $[_{DP}$ the inspection of Beccy$_i$'s office].

Lebeaux (1988, 1991): adjuncts, which include RCs, can be merged late; complements can't (34).

(34) a. $[_{CP}\ [_{DP}$ What claim $[_{RC}$ that B.$_i$ made]] did $[_{TP}$ she$_i$ believe $[_{DP}$ what claim]]]
✓ RC merged late ⇒ no Condition C

b. $[_{CP}\ [_{DP}$ Which inspection of B.$_i$'s office] did $[_{TP}$ she$_i$ resent $[_{DP}$ which $[_{R}$ inspection of B.$_i$'s office]]]]
✗ complement *in situ* ⇒ Condition C

NP complements (Takahashi and Hulsey 2009; Gong 2022). There are exceptions to the ban on late merging complements. When a DP raises out of a non-finite clause, an NP-complement can escape Condition C (35). This follows if D raises alone, and the whole NP-restrictor, is merged above *her* (36).

(35) $[_{DP}$ Every inspection of Beccy$_i$'s office] seems to her$_i$ __ to be unnecessary.

(36) a. $[_{TP}$ seems to her$_i$ $[_{TP}$ $[_{DP}$ every] to be unnecessary]]
D merged alone

b. [$_{TP}$ [$_{DP}$ every [$_{NP}$ inspection of B.$_i$'s office]] seems to her$_i$ [$_{TP}$ [$_{DP}$ every] to be unnecessary]]
NP-restrictor

Thus, both adjuncts and complements can, in principle, be merged counter-cyclically. (32b) must be bad for independent reasons (see §4.5).

Interpreting bare D copies. Given that the R of a DP can be merged counter-cyclically, the question is how do lower bare D copies *without* an R get interpreted in LF?

Following Fox (1999, 2002), I assume that lower DP copies are interpreted via the LF mechanism *Trace Conversion*, which (i) adds a predicate $\lambda y.\ y = x$ to the copy's restrictor, and (ii) replaces its D with THE. This turns any lower copy into a definite description of type e, bound by the higher copy: 'the R that is x'.

To see how this works, consider the toy example in (37). Here, the lower copy of *which boy* becomes 'the boy that is x', a type-e expression bound by the higher copy, while R (*boy*) is still interpreted low:

(37) Which boy did Mary visit __?
a. LF: [$_{DP}$ which boy] λx Mary visited [$_{DP}$ which boy]
b. Insert a variable: [$_{DP}$ which [boy $\lambda y.\ y = x$]]
c. Replace D with THE: [$_{DP}$ the [boy $\lambda y.\ y = x$]] 'the boy that is x'
d. Final LF: [$_{DP}$ which boy] λx Mary visited [$_{DP}$ the [boy $\lambda y.\ y = x$]]
e. Interpretation: which x [Mary visited the boy that is x]

Takahashi and Hulsey (2009) propose that when the lower copy is a bare D, Trace Conversion still applies, but there is no R downstairs for $\lambda y.\ y = x$ to combine with, so the output is just the variable x:

(38) [$_{DP}$ D] $\Rightarrow$ [$_{DP}$ the [$\lambda y.\ y = x$]] $= x$ just a variable, type e

For Takahashi and Hulsey (2009), bare D copies *always* undergo Trace Conversion, so they can never be reconstruction sites: the lower copy is type e, so the λ-abstract over it is a type $\langle e, t\rangle$ predicate, and the type $\langle\langle e, t\rangle, t\rangle$ DP must take it as its argument. (this will be revised later.)

(39) [$_{DP}$ D R] λx [… Y … x …] $\Rightarrow$ $\underbrace{[\![\text{D R}]\!]}_{\langle\langle e,t\rangle,t\rangle}\big(\underbrace{\lambda x\,[\ldots[\![\text{Y}]\!]\ldots x\ldots]}_{\langle e,t\rangle}\big)$ D > Y

So D can only scope where R is. This derives English scope trapping (4a): to avoid Condition C, R is merged above *her*; the lower copy is then a bare D, i.e. just x, so only $\exists >$ *seem*:

(40) [$_{TP}$ [$_{DP}$ a [$_{R}$ student of B.$_i$'s]] seems to her$_i$ [$_{TP}$ [$_{DP}$ a] to be highly intelligent]] $\exists >$ *seem* only

To summarize: DPs can be base-generated as a bare D and can have their R merged *in the course of the derivation*. In the next section, I am going to show that the timing of when R merges can capture the Japanese scope, binding, and Condition C facts with scrambled DOs.

4.3 PROPOSAL: PHASE-TIMED LATE MERGER

I am going to sketch a proposal that captures our Japanese facts (G1–G3). The proposal has three ingredients, concerning (i) *when* R is merged to the scrambled DO; (ii) *how* the copies in a movement chain get converted in LF to an interpretable representation; and (iii) *where* D and R get interpreted.

Step 1: when is R merged? I propose that, quite generally, Rs are merged late: they can merge with the highest DP copy upon completion of a phase (vP and CP).

(41) **Merge R:** R can be merged at the completion of a phase on D's path, on the *highest* copy of D at that point. If D has not yet moved, that copy is its base position.

There's good reason to think that if counter-cyclic merger is an option, then it must be taken. As shown in (42a), (43a), a reciprocal inside an RC cannot be bound by the subject. This indicates that the RC must late-adjoin to the DP (and cannot adjoin in the base position) (Stepanov 2001a,b).

(42) a. ?* What evidence that each other$_i$'s friends brought up at court did the lawyers$_i$ refuse to talk about?
b. The lawyers$_i$ refused to talk about the evidence that each other$_i$'s friends brought up at court.

(43) a. ?* Which proofs that demonstrated that pictures of each other$_i$ had been forged did they$_i$ read?
b. They$_i$ read proofs that demonstrated that pictures of each other$_i$ had been forged.

(Stepanov 2001b: 100)

If Condition A needs c-command at some point in the derivation, the reciprocal in (42a), (43a) was never c-commanded by its antecedent. So the relative clause was adjoined only after *wh*-movement: adjunction is obligatorily late.

As for how late, Zyman (2022) gives compelling evidence that counter-cyclic merger occurs upon completion of the phase (vP, CP).

Exactly can modify an *in situ wh*-phrase (44). It cannot be stranded VP-internally when *what* moves (45), but it can be stranded pre-verbally, at the vP edge (46):

(44) a. Who put what exactly on the table with great care?
b. Who gave what exactly to the president to annoy you?

(Zyman 2022: 90)

(45) a. ?* What did Muriel put exactly on the table with great care?
b. ?* What did Harvey give exactly to the president to annoy you?

(Zyman 2022: 90)

(46) a. What did she exactly send?
b. What did they exactly change in the newest update?

(Zyman 2022: 100)

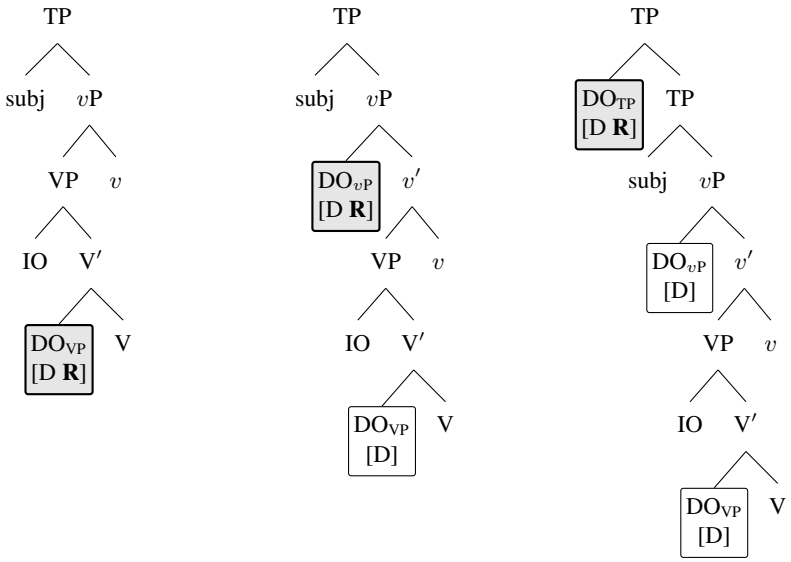
Zyman's account: *exactly* adjoins only once the vP phase is complete, and cannot adjoin to a lower copy. By then the *wh*-phrase has moved to Spec,vP, so there is nothing left in VP to adjoin to:

(47) a. [$_{vP}$ *wh*-XP$_i$ v [$_{VP}$ V t$_i$]] vP complete
b. *[$_{vP}$ *wh*-XP$_i$ v [$_{VP}$ V [t$_i$ exactly]]] no adjunction to a lower copy

(48) a. [$_{vP}$ *wh*-XP$_i$ v [$_{VP}$ V t$_i$]] vP complete
b. [$_{vP}$ [*wh*-XP$_i$ exactly] v [$_{VP}$ V t$_i$]] late adjunction at the edge
c. [$_{CP}$ *wh*-XP$_i$ C … [$_{vP}$ [t$_i$ exactly] v [$_{VP}$ V t$_i$]]] *wh*-movement strands *exactly*

Returning to Japanese scrambling: if R can only be merged on the highest copy when vP or CP is completed, there are three possible merge sites:

(49)



(a) DO not scrambled: R merged in VP (ordinary merge)

(b) DO scrambled: R merged at the vP edge at vP completion

(c) R not merged at vP: R merged in TP at CP completion

Step 2: how are the copies in a chain interpreted? When a phase (vP, CP) is completed, it is sent to the interfaces. In LF, the DP chain inside it has to be converted into something the semantics can interpret.

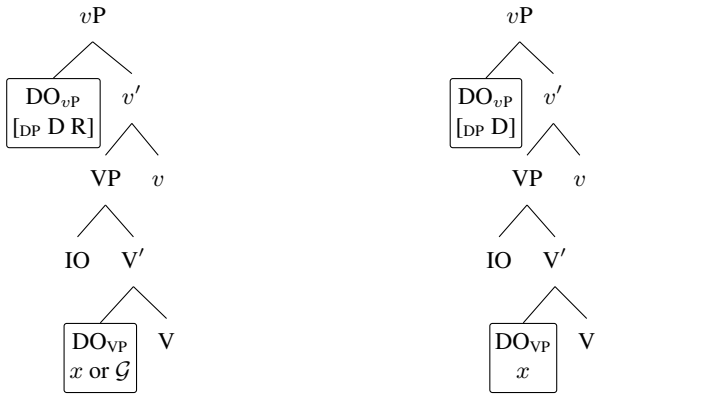
A lower copy *with* R is converted by ordinary Trace Conversion (37). For a lower copy *without* R, I suggest there are two options:

(50) **Interpret copies:** at phase completion, an R-less copy of D is interpreted as

a. a bare variable x of type e, via Trace Conversion: **always** available; or

b. a higher-type trace of type $\langle\langle e,t\rangle,t\rangle$, matching the type of [$_{DP}$ D R]: available **only if** the chain already contains R.

To illustrate, compare two snapshots of the vP phase at the point it is completed. On the left, R has already been merged on the copy at the vP edge, so the R-less lower copy can be converted to a variable or a higher-type trace. On the right, R hasn't been merged yet, so the lower copy can only be a variable:



R at the vP edge: copy in VP = x *or* $\mathcal{G}$ no R yet: copy in VP = x only

Motivation for (50b). A higher-type trace has the type of its binder (Cresti 1995; Rullmann 1995). If the chain already has R when the phase is sent to LF, the binder is a full DP, of type $\langle\langle e,t\rangle,t\rangle$, so the lower copy can match it. If the chain doesn't have R yet, the binder is a bare D, of type $\langle\langle e,t\rangle,\langle\langle e,t\rangle,t\rangle\rangle$: a trace of that type would need an R beside it, and there is none, so Trace Conversion *must* apply (Takahashi and Hulsey 2009). Since an interpreted phase isn't revisited, merging R later can't retype the trace.

Step 3: what determines scope, and what determines binding and Condition C? The DP is interpreted as a unit; R is never interpreted separately from D. But scope and binding/Condition C are sensitive to different things:

(51) a. **Scope:** the DP takes scope in any copy that hasn't been converted to a variable: either the highest copy, or a lower copy that's been converted to a higher-type trace.
b. **Binding and Condition C:** these are checked on the LF structure, so what matters is where R sits in it: the lowest copy containing R, i.e. its merge position.

The key point, when we combine Steps 2 and 3:

- The DP can always take scope **above** R's position: R stays in its merge position in the LF structure, while the DP can take scope from a higher copy.
- The DP can take scope **below** R's position only via a higher-type trace, and a higher-type trace requires R to be in the chain when the phase is interpreted (50b). So the DP can scope below R's position only within the phase where R was merged.

(52) a. **Merge R** upon completion of a phase (vP/CP), on the highest copy of D. (41)
b. **Interpret copies** at phase completion: an R-less copy can always be a variable; it can also be a higher-type trace, but only if the chain already contains R. (50)
c. **Scope** is taken in any copy that isn't a variable; **binding and Condition C** see R in its merge position. (51)

4.4 Deriving G1–G3

With all this machinery, we are now in a position to see how G1–G3 can be explained.

G1: a DO scrambled to vP

Recall that a DO scrambled over the IO can scope below it (53a), but can't be bound by it (53b):

(53) a. John-wa [$_{DO}$ dareka-o] dono sensee-ni-mo __ syookaisita.
John-TOP someone-ACC which teacher-DAT-MO introduced
'John introduced someone to every teacher.' ✓ $\forall > \exists$ = (12a)
b.?? John-wa [$_{DO}$[$_{RC}$ soko$_1$-ni funinsuru] sensee]-o [$_{IO}$ dono gakkoo-ni-mo$_1$] __ syookaisita.
John-TOP there-DAT be.assigned teacher-ACC which school-DAT-MO introduced
Int. 'John introduced to every school$_1$ a teacher assigned there$_1$.' ✗ bound = (13b)

Importantly, the DO can scope below a pronominal IO without triggering Condition C, i.e. there's no scope trapping at the IO:

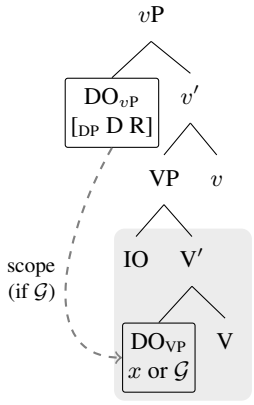
(54) John-wa [$_{DO}$ [Ken-to Lee-to Mia]$_i$-no tomodati]-o karera$_i$ minna-ni __ syookaisita.
John-TOP Ken-and Lee-and Mia-GEN friend-ACC they all-DAT introduced
'John introduced a friend of Ken, Lee and Mia$_i$ to them$_i$ all.' ✓ coreference + $\forall > \exists$ = (15b)

Step 1: merge R. When vP is completed, the DO is already at the vP edge. Per (41), R is merged there, above the IO.

Step 2: interpret copies. When vP is sent to LF, the chain already contains R. So, per (50), the R-less copy in VP can be either a variable or a higher-type trace.

Step 3: scope, binding, Condition C. In the LF structure, R sits at the vP edge, outside the IO's c-command domain: so the IO can't bind into it, and an IO pronoun doesn't trigger Condition C. For scope, the DP can scope at the vP edge (copy in VP = x), or below the IO (copy in VP = $\mathcal{G}$):

(55)



1. vP completed: R merged on DO$_{vP}$
2. DO$_{VP}$ = x or $\mathcal{G}$
3. R outside the IO's domain;
   DO$_{vP}$ scopes over IO or, via $\mathcal{G}$, reconstructs below to VP.

This gives us G1:

- The DP can scope below the IO, via $\mathcal{G}$ (53a), or above it, via x.
- R is outside the IO's c-command domain, so the IO can't bind into it (53b).
- Since the DP can scope low while R stays outside the IO's domain, there's no scope trapping at the IO (54).

G2: a DO scrambled to TP can't scope below R's position

Recall that with a full-NP subject, a DO scrambled to TP can scope below the IO (56a); with a coreferent subject pronoun, it can't (56b):

(56) a. [$_{DO}$ ni-mai-no John-no nigaoe]-o Mary-ga dono gakusei-ni-mo __ miseta (koto)
two-CL-GEN John-GEN portrait-ACC Mary-NOM which student-DAT-MO showed fact
'Mary showed every student two portraits of John.' ✓ $2 > \forall$; ✓ $\forall > 2$ = (20b)

b. [$_{DO}$ ni-mai-no John$_i$-no nigaoe]-o kare$_i$-ga dono gakusei-ni-mo __ miseta (koto)
two-CL-GEN John-GEN portrait-ACC he-NOM which student-DAT-MO showed fact
'He$_i$ showed every student two portraits of John$_i$.' ✓ $2 > \forall$; ✗ $\forall > 2$ = (22b)

As we saw, (56b) disallows scope reconstruction because of Condition C: scope trapping.

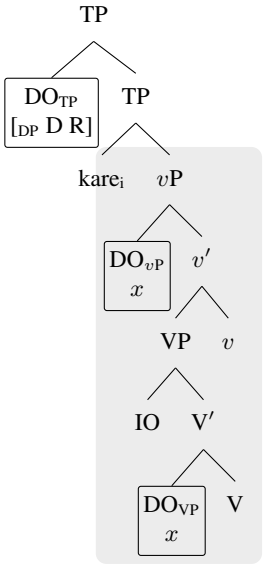
Step 1: merge R. R can't be merged at vP completion: the vP edge is inside *kare*'s c-command domain, so *John* would trigger Condition C. So R is merged in TP, at CP completion. When vP is sent to LF, the chain has only a bare D.[2]

Step 2: interpret copies. Since the chain has no R when vP is sent to LF, the copies in VP and at the vP edge can only be variables. Merging R in a later cycle can't change this.

Step 3: scope, binding, Condition C. R sits in TP, outside *kare*'s c-command domain: no Condition C. Since all the lower copies are variables, there's no higher-type trace for the DP to be fed into, so it can only scope in TP. That's scope trapping:

[2]This raises a look-ahead question: how does the derivation "know" at vP completion that the vP edge will be c-commanded by *kare*? One option is that vP isn't sent to the interfaces until C is merged.

(57)



1. vP completed: no R yet
2. DO$_{VP}$, DO$_{vP}$ = x only
3. CP completed: R merged on DO$_{TP}$
4. R outside *kare*'s domain;
   DO$_{TP}$must take wide scope

This gives us G2:

- Coreference is fine, with the DP scoping high: $2 > \forall$ in (56b).
- The DP can't scope below the subject or the IO: ✗ $\forall > 2$ in (56b).
- With a non-pronominal subject (56a), nothing blocks merging R at the vP edge. Then the chain has R when vP is sent to LF, the copy in VP can be $\mathcal{G}$, and $\forall > 2$ is available.

G3: a DO scrambled to TP can have R below the subject

Recall that the subject can bind into a scrambled DO that scopes over it:

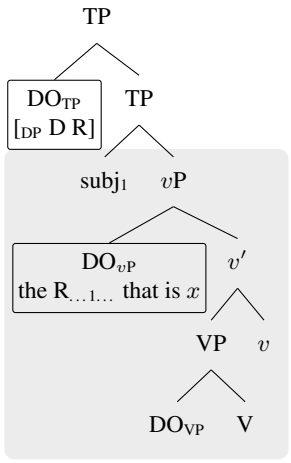
(58) [$_{DO}$[$_{RC}$ Soitu$_1$-ga kai-ta] aru ronbun]-o [$_S$ san-nin-ika-no kooho-sya-ga]$_1$
that.guy-NOM write-PST certain paper-ACC three-CL-or.fewer-GEN candidate-person-NOM
__ teisyutu-si-ta.
submission-do-PST
'A certain paper he$_1$ wrote, at most three candidates$_1$ submitted.' ✓ bound; *aru* > subj = (25)

Step 1: merge R. R is merged at vP completion, at the vP edge, below the subject. The DO, now with R, moves on to TP.

Step 2: interpret copies. The copy at the vP edge contains R, so it undergoes ordinary Trace Conversion: 'the R that is x'.

Step 3: scope, binding, Condition C. In the LF structure, R is present down to the vP edge, inside the subject's c-command domain: so the subject can bind into it (51b). For scope, the DP scopes from the highest copy, in TP, above the subject (51a):

(59)



1. vP completed: R merged on DO$_{v\text{P}}$
2. DO moves on to TP, with R
3. R inside the subject's domain: bound
4. DO$_{\text{TP}}$scopes high above the subject

This gives us G3: the DO is bound by the subject and scopes over it.
In short: the merge position of R determines both (i) where R sits in the LF structure, for binding and Condition C, and (ii) whether the DP can scope low.

If R is merged at the vP edge, the chain has R when vP is interpreted, so the DP can scope below R's position via a higher-type trace. But if R has to wait until TP to merge (to avoid Condition C), the lower copies have already been fixed as variables, and the DP can only scope high.

4.5 TO MANY MERGE POSITIONS?

As currently stated, late merge upon completion of the vP/CP phase, overgenerates for English.

E.g. a *wh*-object passes through the vP edge, so its NP-restrictor could be merged above the IO and the subject, and (60) should escape Condition C (cf. the RC-restrictor in (61)):

(60) *[$_{DP}$ Which [$_{NP}$ picture of Peter$_i$]] did Martha show him$_i$ __?

(61) [$_{DP}$ Which article [$_{RC}$ that Peter$_i$ wrote]] did Martha give him$_i$ __?

(62) If the DP is Case-licensed under AGREE, the NP-restrictor must be present at the point where the DP is licensed. (cf. Takahashi and Hulsey 2009)

Takahashi and Hulsey (2009) propose that Case is a requirement on the entire nominal, i.e. D + NP-restrictor, so the entire nominal must be present when the DP undergoes AGREE with T/v. Since v assigns ACC to the DO in (60) while the DP is inside VP, the NP-restrictor must be merged with D in the base position. Hence the Condition C effect in (60). In contrast, since RCs are optional adjuncts, they can always be merged late (Lebeaux 1988, 1991).

What if Japanese doesn't have (62)? Japanese lacks ϕ-agreement (Kuroda 1988), and its case particles are not reflexes of Agree with v or T. Instead, a case particle is a K head that projects above DP and is present throughout the derivation

(63) [$_{KP}$ [$_{DP}$ ni-mai [$_{R}$ John-no nigaoe]] -o]

I suggest that the absence of Case-licensing under AGREE is what lifts the cap on when the NP-restrictor can merge in Japanese.

5 CONCLUSION

- In English, scope, binding, and Condition C go together. In Japanese scrambling, D and R can be interpreted in different copies (G1, G3), but not freely (G2).
- This is very different from English reconstruction effects where scope, binding, and Condition C all move together.

- This is quite tricky to capture. However, I've outlined an approach that I think is on the way there.

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